Supplementary Material for:

Knowledge-Guided Dual-Phase Swarm Evolutionary Learning for High-Dimensional Feature Selection

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Abstract

This supplementary document provides comprehensive extended experimental records, complete per-dataset metrics, detailed hyper-parameter sensitivity grids, convergence trajectories, and full statistical significance matrices supporting the main manuscript.

Table of Contents

- **Section S1**: Benchmark Dataset Characteristics and Selection Rationale
- **Section S2**: Baseline Algorithms and Detailed Hyper-Parameter Configurations
- **Section S3**: Complete Per-Dataset Classification Accuracy and Feature Compactness Metrics
- **Section S4**: Granular Per-Dataset Component Contribution (Ablation) Results (**Table S4**)
- **Section S5**: Granular Per-Dataset Computational Wall-Clock Runtime Measurements (**Table S5**)
- **Section S6**: Full Multi-Dataset Parameter Sensitivity Evaluation Grids (**Table S6**)
- **Section S7**: Full Convergence Trajectories and Boxplot Distributions
- **Section S8**: Extended Statistical Significance Testing Matrices

Section S1: Benchmark Dataset Characteristics and Selection Rationale

Table 2 summarizes the statistical characteristics of the 20 benchmark datasets investigated in this study, spanning biomedical gene-expression microarrays, high-dimensional text categorization, visual object recognition, and acoustic phoneme analysis.

Table 1:

Table 2: Dataset Description (20 Benchmark Datasets)

Dataset	Samples	Features	Classes	Domain
Adenoma	36	7,457	2	Gene expression
ALL_AML	72	7,129	2	Gene expression
ALL1	128	12,625	2	Gene expression
ALL4	93	12,625	2	Gene expression
Colon	62	2,000	2	Gene expression
DLBCL	77	7,129	2	Gene expression
Gastric	65	22,645	2	Gene expression
Gastric1	144	22,283	2	Gene expression
Gastric2	124	22,283	2	Gene expression
Leukaemia	72	7,129	2	Gene expression
Lymphoma	45	4,026	2	Gene expression
Myeloma	173	12,625	2	Gene expression
Prostate	102	12,625	2	Gene expression
SRBCT	83	2,308	4	Gene expression
T1D	101	54,675	2	Gene expression
BASEHOCK	1,993	4,862	2	Text
COIL20	1,440	1,024	20	Image
ISOLET	1,560	617	26	Speech
madelon	2,600	500	2	Synthetic
PCMAC	1,943	3,289	2	Text

Range: $n \in [36, 2600]$, $d \in [500, 54675]$, $c \in [2, 26]$

Section S2: Baseline Algorithms and Detailed Hyper-Parameter Configurations

Table 4 lists the full taxonomy and categorization of the 13 baseline algorithms evaluated in the comparative study, and Table 6 specifies the precise hyper-parameter configurations used across all algorithms to ensure strict reproducibility.

Table 3:

Table 4: The 14 fixed-length methods compared in this study.

Method	Category	Reference
KDS-EL	Proposed	This work
KDS-EL w/o TPS	Ablation	This work
KDS-EL w/o TPS&KER	Ablation	This work
MEL	Baseline	[1]
FWPSO	Single-objective	[2]
PSO-CSM	Single-objective	[3]
SaWDE	Single-objective	[4]
IGPSO	Single-objective	[5]
BPSO	Single-objective	[6]
W-MOPSO-FS	Multi-objective	Self-built
fMOPSO-FS	Multi-objective	[7]
NSGA-II-FS	Multi-objective	[8]
MOEA-D-FS	Multi-objective	[9]
MOPSO-FS	Multi-objective	[10]

Section S3: Complete Per-Dataset Classification Accuracy and Feature Compactness Metrics

Tables 7 and 8 present the complete per-dataset 5-fold cross-validation classification accuracy (mean ± standard deviation) and average selected feature counts across all 20 benchmark datasets and all 14 evaluated algorithms.

Table 5:

Table 6: Parameter Settings

Common Experimental Setup		
Population size (N)	=	20
Maximum iterations (T)	=	100
KNN neighbors (k)	=	3
Real-to-binary threshold (θ)	=	0.6
Cross-validation	=	5-fold stratified, shuffled
Number of independent runs	=	10
KDS-EL (full framework)		
Inertia weight (w)	=	0.9
Acceleration coefficients	=	$c_1 = c_2 = c_3 = 2.0$
Knowledge weight decay	=	0.99
Elite flip fraction	=	0.01
Elite count (k_e)	=	3
Random flip probability	=	0.10
MI-guided init ratio	=	0.5
Stall patience (P_s)	=	20
Phase split ratio	=	0.7
Phase-1 objective weights	=	$\mathbf{w}_s = [0.7, 0.3]$
Phase-2 objective weights	=	$\mathbf{w}_s = [0.95, 0.05]$
KDS-EL w/o TPS (no two-phase scheduling)		
Fixed objective weights	=	$\mathbf{w}_s = [0.9, 0.1]$
MEL and KDS-EL w/o TPS&KER		
Inertia weight (w)	=	0.9
Acceleration coefficients	=	$c_1 = c_2 = c_3 = 2.0$
MI-guided init ratio (KDS-EL w/o TPS&KER)	=	0.5
Stall patience (KDS-EL w/o TPS&KER)	=	20
Competitor Methods		
Competitors (BPSO, FWPSO, IGPSO, PSO-CSM, SaWDE, NSGA-II-FS, MOPSO-FS, MOEA-D-FS, fMOPSO-FS, W-MOPSO-FS) use the hyper-parameters recommended in their original publications, with $N = 20$ and $T = 100$ unless the original paper specifies otherwise.		

Section S4: Granular Per-Dataset Component Contribution (Ablation) Results (Table S4)

Table 9 provides the complete per-dataset performance metrics across the four sequential configurations of the subtractive ablation chain: (i) MEL baseline [1], (ii) MEL+MI (w/o TPS&KER), (iii) KDS-EL$_{\text{KER}}$ (w/o TPS), and (iv) Complete KDS-EL.

Table 7: Classification accuracy (mean ± std) of 14 fixed-length methods on 20 benchmark datasets. Bold indicates the best per dataset; +, ≈, and − in parentheses report two-sided Wilcoxon signed-rank tests ($p < 0.05$) over the 10 runs of each dataset between KDS-EL and the column method: + KDS-EL significantly better, ≈ no significant difference, − KDS-EL significantly worse. The Average row reports the mean across the 20 datasets.

Dataset	KDS-EL	KDS-EL w/o TPS	KDS-EL w/o TPS&KER	MEL	FWPSO	PSO-CSM	SaWDE	IGPSO	BPSO	W-MOPSO-FS	fMOPSO-FS	NSGA-II-FS	MOEA-D-FS	MOPSO-FS
ALL1	0.9998±0.0001	0.9996±0.0000(+)	0.9996±0.0000(+)	0.9664±0.0023(+)	0.9952±0.0015(+)	0.9900±0.0002(+)	0.9710±0.0011(+)	0.9802±0.0002(+)	0.9534±0.0003(+)	**1.0000±0.0000**(−)	**1.0000±0.0000**(−)	**1.0000±0.0000**(−)	0.9985±0.0031(≈)	**1.0000±0.0000**(−)
ALL4	**0.9522±0.0061**	0.9418±0.0113(≈)	0.9415±0.0112(+)	0.8704±0.0661(+)	0.8617±0.0304(+)	0.8363±0.0135(+)	0.8062±0.0217(+)	0.7914±0.0179(+)	0.7569±0.0069(+)	0.8068±0.0047(+)	0.8015±0.0089(+)	0.7694±0.0118(+)	0.7737±0.0184(+)	0.7654±0.0074(+)
ALL_AML	**0.9995±0.0002**	0.9990±0.0002(+)	0.9886±0.0016(+)	0.9539±0.0008(+)	0.9850±0.0067(+)	0.9748±0.0039(+)	0.9591±0.0012(+)	0.9669±0.0003(+)	0.9426±0.0034(+)	0.9943±0.0070(≈)	0.9900±0.0065(+)	0.9857±0.0000(+)	0.9857±0.0000(+)	0.9857±0.0000(+)
Adenoma	**0.9996±0.0002**	**0.9996±0.0001**(≈)	0.9958±0.0081(+)	0.9410±0.0011(+)	0.9783±0.0143(+)	0.9665±0.0108(+)	0.9461±0.0014(+)	0.9527±0.0003(+)	0.9287±0.0002(+)	0.9800±0.0131(+)	0.9743±0.0086(+)	0.9714±0.0000(+)	0.9714±0.0000(+)	0.9714±0.0000(+)
BASEHOCK	0.9113±0.0120	0.8997±0.0114(+)	0.8597±0.0062(+)	0.8518±0.0063(+)	0.8715±0.0095(+)	0.8637±0.0050(+)	0.8860±0.0065(+)	0.8479±0.0062(+)	0.8597±0.0036(+)	0.8847±0.0066(+)	0.8861±0.0045(+)	**0.9238±0.0082**(−)	0.9150±0.0086(≈)	0.8920±0.0034(+)
COIL20	0.9939±0.0011	0.9919±0.0008(+)	0.9619±0.0007(+)	0.9634±0.0017(+)	0.9894±0.0018(+)	0.9782±0.0006(+)	0.9763±0.0017(+)	0.9707±0.0010(+)	0.9573±0.0006(+)	0.9960±0.0009(−)	0.9960±0.0005(−)	**0.9971±0.0007**(−)	0.9958±0.0010(−)	0.9964±0.0009(−)
Colon	**0.9500±0.0075**	0.9347±0.0094(+)	0.9211±0.0092(+)	0.8548±0.0276(+)	0.9221±0.0203(+)	0.8880±0.0090(+)	0.8867±0.0110(+)	0.8636±0.0101(+)	0.8529±0.0059(+)	0.9033±0.0072(+)	0.8988±0.0071(+)	0.8731±0.0097(+)	0.8745±0.0145(+)	0.8806±0.0104(+)
DLBCL	**0.9995±0.0002**	0.9993±0.0004(≈)	0.9902±0.0016(+)	0.9483±0.0058(+)	0.9819±0.0092(+)	0.9819±0.0048(+)	0.9629±0.0055(+)	0.9655±0.0063(+)	0.9435±0.0069(+)	0.9974±0.0052(≈)	0.9960±0.0061(≈)	0.9781±0.0060(+)	0.9781±0.0060(+)	0.9872±0.0058(+)
Gastric	**0.9782±0.0034**	0.9677±0.0093(+)	0.9748±0.0088(≈)	0.8955±0.0786(+)	0.9191±0.0340(+)	0.8874±0.0087(+)	0.8234±0.0191(+)	0.8564±0.0087(+)	0.8046±0.0093(+)	0.8908±0.0046(+)	0.8846±0.0077(+)	0.8154±0.0069(+)	0.8169±0.0083(+)	0.8215±0.0102(+)
Gastric1	**0.9800±0.0003**	0.9738±0.0004(+)	0.9642±0.0025(+)	0.9310±0.0155(+)	0.9597±0.0032(+)	0.9496±0.0024(+)	0.9348±0.0030(+)	0.9402±0.0018(+)	0.9145±0.0002(+)	0.9588±0.0021(+)	0.9581±0.0000(+)	0.9588±0.0021(+)	0.9554±0.0034(+)	0.9588±0.0021(+)
Gastric2	**0.9981±0.0029**	0.9936±0.0029(+)	0.9872±0.0038(+)	0.9522±0.0170(+)	0.9837±0.0049(+)	0.9710±0.0002(+)	0.9532±0.0038(+)	0.9623±0.0001(+)	0.9357±0.0034(+)	0.9837±0.0000(+)	0.9838±0.0002(+)	0.9765±0.0025(+)	0.9765±0.0025(+)	0.9832±0.0025(+)
ISOLET	0.8894±0.0056	0.8898±0.0090(≈)	0.8557±0.0063(+)	0.8501±0.0067(+)	0.8822±0.0109(≈)	0.8584±0.0043(+)	0.8903±0.0076(≈)	0.8468±0.0070(+)	0.8607±0.0027(+)	0.8883±0.0032(≈)	0.8874±0.0066(≈)	**0.9090±0.0059**(−)	0.8963±0.0059(−)	0.8828±0.0036(+)
Leukaemia	**0.9995±0.0002**	0.9993±0.0002(≈)	0.9895±0.0017(+)	0.9534±0.0015(+)	0.9886±0.0059(+)	0.9775±0.0059(+)	0.9587±0.0017(+)	0.9682±0.0037(+)	0.9416±0.0002(+)	0.9886±0.0057(+)	0.9886±0.0057(+)	0.9857±0.0000(+)	0.9857±0.0000(+)	0.9857±0.0000(+)
Lymphoma	**0.9994±0.0003**	**0.9994±0.0004**(≈)	0.9868±0.0102(+)	0.8731±0.0239(+)	0.9404±0.0306(+)	0.9206±0.0098(+)	0.8841±0.0092(+)	0.8861±0.0139(+)	0.8585±0.0081(+)	0.9178±0.0102(+)	0.9200±0.0109(+)	0.8844±0.0133(+)	0.8867±0.0067(+)	0.8911±0.0067(+)
Myeloma	**0.9155±0.0101**	0.8971±0.0072(+)	0.8825±0.0084(+)	0.8450±0.0244(+)	0.8702±0.0069(+)	0.8550±0.0079(+)	0.8522±0.0071(+)	0.8304±0.0077(+)	0.8186±0.0056(+)	0.8535±0.0037(+)	0.8529±0.0060(+)	0.8594±0.0070(+)	0.8482±0.0115(+)	0.8425±0.0035(+)
PCMAC	**0.8705±0.0139**	0.8424±0.0163(+)	0.8007±0.0051(+)	0.7870±0.0080(+)	0.8103±0.0090(+)	0.8036±0.0095(+)	0.8256±0.0095(+)	0.7840±0.0058(+)	0.7982±0.0062(+)	0.8175±0.0056(+)	0.8105±0.0086(+)	0.8523±0.0083(+)	0.8505±0.0117(+)	0.8169±0.0035(+)
Prostate	**0.9701±0.0032**	0.9595±0.0045(+)	0.9638±0.0049(+)	0.8458±0.0083(+)	0.8887±0.0092(+)	0.8721±0.0088(+)	0.8711±0.0091(+)	0.8526±0.0102(+)	0.8485±0.0038(+)	0.8840±0.0031(+)	0.8839±0.0054(+)	0.8704±0.0120(+)	0.8703±0.0107(+)	0.8743±0.0052(+)
SRBCT	0.9986±0.0004	0.9984±0.0006(≈)	0.9944±0.0043(+)	0.9321±0.0085(+)	0.9692±0.0148(+)	0.9695±0.0091(+)	0.9659±0.0048(+)	0.9515±0.0116(+)	0.9488±0.0050(+)	**1.0000±0.0000**(−)	**1.0000±0.0000**(−)	0.9853±0.0074(+)	0.9829±0.0099(+)	0.9829±0.0080(+)
T1D	**0.8598±0.0094**	0.8232±0.0126(+)	0.8246±0.0084(+)	0.7679±0.0461(+)	0.7610±0.0169(+)	0.7318±0.0065(+)	0.7237±0.0049(+)	0.7113±0.0064(+)	0.7063±0.0041(+)	0.7256±0.0049(+)	0.7227±0.0053(+)	0.7118±0.0063(+)	0.7088±0.0088(+)	0.7179±0.0048(+)
madelon	**0.8822±0.0180**	0.8659±0.0246(≈)	0.7679±0.0121(+)	0.7521±0.0095(+)	0.7963±0.0165(+)	0.7669±0.0229(+)	0.8106±0.0089(+)	0.7497±0.0169(+)	0.7652±0.0071(+)	0.7920±0.0049(+)	0.7845±0.0048(+)	0.8214±0.0097(+)	0.8122±0.0055(+)	0.7808±0.0071(+)
Average	**0.9574**	0.9488	0.9325	0.8868	0.9177	0.9021	0.8944	0.8839	0.8698	0.9132	0.9110	0.9065	0.9042	0.9009

Table 8: Average number of selected features (over 10 runs) of 14 methods on 20 datasets. Bold indicates the minimum per dataset.

Dataset	KDS-EL	KDS-EL w/o TPS	KDS-EL w/o TPS&KER	MEL	FWPSO	PSO-CSM	SaWDE	IGPSO	BPSO	W-MOPSO-FS	fMOPSO-FS	NSGA-II-FS	MOEA-D-FS	MOPSO-FS
ALL1	**39**	55	55	4248	612	1206	3667	2501	5886	2333	2339	5368	5548	5920
ALL4	124	152	**64**	1082	1326	1447	4581	2677	6217	3645	3213	5461	5736	6126
ALL_AML	**67**	70	473	2369	521	949	1996	1444	3266	1678	1562	2872	2968	3266
Adenoma	66	**33**	120	2483	278	937	2102	1608	3399	1691	1551	3000	3127	3420
BASEHOCK	**134**	208	812	1965	2066	1113	1912	1894	2419	1390	1617	2073	2150	2319
COIL20	84	**60**	348	323	90	167	232	255	407	405	407	283	286	405
Colon	**14**	35	105	702	240	284	621	448	935	580	612	690	730	903
DLBCL	64	**52**	615	2784	876	1074	2155	1718	3379	1963	2067	2904	3010	3349
Gastric	100	97	**62**	3604	1070	2782	7072	4916	11004	5744	6082	10165	10342	10938
Gastric1	**31**	217	678	7393	1420	2966	6963	5074	10661	6148	6044	9954	10128	10706
Gastric2	**83**	140	444	7084	1362	3062	6708	5134	10627	5803	5956	9919	10062	10701
ISOLET	**60**	100	225	250	172	194	213	195	272	242	255	185	174	256
Leukaemia	66	**47**	488	2409	534	934	2026	1440	3246	1492	1503	2893	2973	3272
Lymphoma	45	**23**	50	1404	304	524	1204	880	1832	1027	1076	1536	1609	1834
Myeloma	250	365	**175**	2137	1863	1847	4543	2859	6259	3854	3866	5585	5839	6113
PCMAC	**91**	329	637	1339	1209	651	1292	1137	1635	966	912	1352	1431	1560
Prostate	26	**23**	27	5058	1898	1541	4229	2729	6211	3294	3384	5532	5676	6100
SRBCT	64	**36**	129	906	431	366	739	515	1083	700	742	818	874	1044
T1D	330	366	**174**	6715	4984	9047	18009	13170	26971	16096	16774	25649	25939	26830
madelon	**18**	26	130	190	122	95	162	171	221	149	174	128	132	204

Table 9: Per-dataset ablation results over the 20-dataset suite: accuracy and mean feature count for each variant of the incremental construction.

Dataset	MEL		MEL+MI		KDS-EL$_{KER}$		KDS-EL	
	Acc	Feat	Acc	Feat	Acc	Feat	Acc	Feat
ALL1	0.9664±0.0023	4248	0.9996±0.0000	55	0.9996±0.0000	55	0.9998±0.0001	39
ALL4	0.8704±0.0661	1082	0.9415±0.0112	64	0.9418±0.0113	152	0.9522±0.0061	124
ALL_AML	0.9539±0.0008	2369	0.9886±0.0016	473	0.9990±0.0002	70	0.9995±0.0002	67
Adenoma	0.9410±0.0011	2483	0.9958±0.0081	120	0.9996±0.0001	33	0.9996±0.0002	66
BASEHOCK	0.8518±0.0063	1965	0.8597±0.0062	812	0.8997±0.0114	208	0.9113±0.0120	134
COIL20	0.9634±0.0017	323	0.9619±0.0007	348	0.9919±0.0008	60	0.9939±0.0011	84
Colon	0.8548±0.0276	702	0.9211±0.0092	105	0.9347±0.0094	35	0.9500±0.0075	14
DLBCL	0.9483±0.0058	2784	0.9902±0.0016	615	0.9993±0.0004	52	0.9995±0.0002	64
Gastric	0.8955±0.0786	3604	0.9748±0.0088	62	0.9677±0.0093	97	0.9782±0.0034	100
Gastric1	0.9310±0.0155	7393	0.9642±0.0025	678	0.9738±0.0004	217	0.9800±0.0003	31
Gastric2	0.9522±0.0170	7084	0.9872±0.0038	444	0.9936±0.0029	140	0.9981±0.0029	83
ISOLET	0.8501±0.0067	250	0.8557±0.0063	225	0.8898±0.0090	100	0.8894±0.0056	60
Leukaemia	0.9534±0.0015	2409	0.9895±0.0017	488	0.9993±0.0002	47	0.9995±0.0002	66
Lymphoma	0.8731±0.0239	1404	0.9868±0.0102	50	0.9994±0.0004	23	0.9994±0.0003	45
Myeloma	0.8450±0.0244	2137	0.8825±0.0084	175	0.8971±0.0072	365	0.9155±0.0101	250
PCMAC	0.7870±0.0080	1339	0.8007±0.0051	637	0.8424±0.0163	329	0.8705±0.0139	91
Prostate	0.8458±0.0083	5058	0.9638±0.0049	27	0.9595±0.0045	23	0.9701±0.0032	26
SRBCT	0.9321±0.0085	906	0.9944±0.0043	129	0.9984±0.0006	36	0.9986±0.0004	64
T1D	0.7679±0.0461	6715	0.8246±0.0084	174	0.8232±0.0126	366	0.8598±0.0094	330
madelon	0.7521±0.0095	190	0.7679±0.0121	130	0.8659±0.0246	26	0.8822±0.0180	18

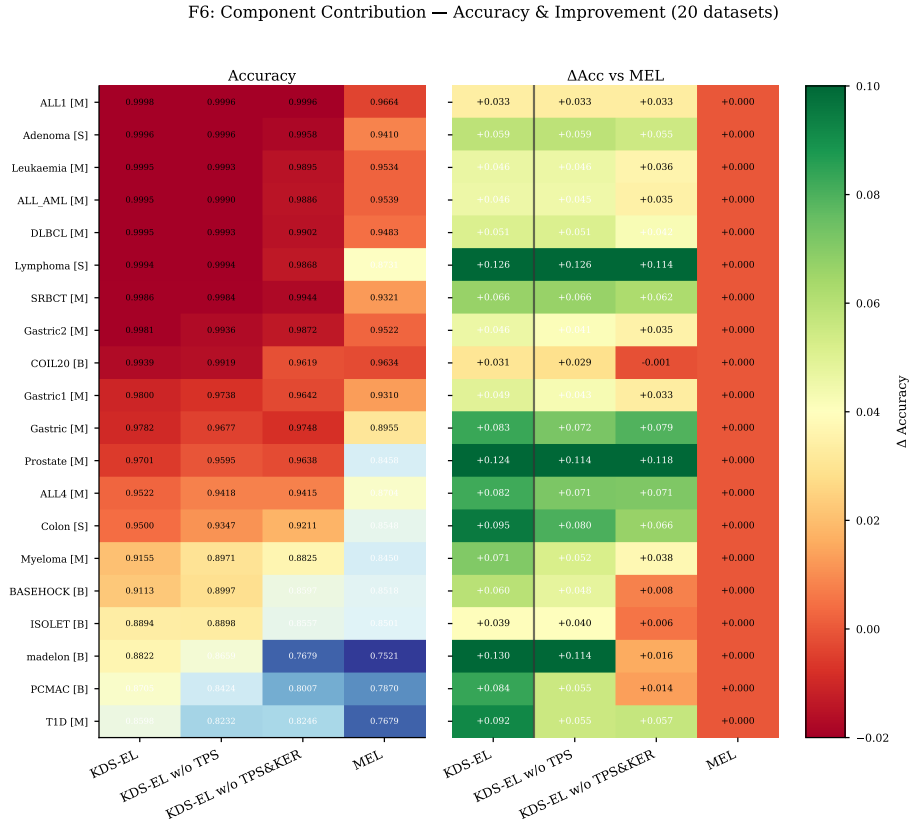


Figure 1: Component contribution heatmap illustrating relative accuracy enhancements and feature reductions across the ablation chain.

Section S5: Granular Per-Dataset Computational Wall-Clock Runtime Measurements (Table S5)

Table 10 details the mean wall-clock computational execution time (in seconds, averaged over 10 independent runs) across all 20 benchmark datasets for the 14 compared algorithms.

Table 10: Per-dataset wall-clock runtime in seconds (mean over 10 runs) for the 14 methods on 20 datasets.

Dataset	KDS-EL	KDS-EL$_{KER}$	MEL+MI	MEL	FWPSO	PSO-CSM	SaWDE	IGPSO	BPSO	W-MOPSO	fMOPSO	NSGA-II	MOEA-D	MOPSO
ALL1	40.9	39.9	39.9	176.0	19.7	82.2	67.6	91.4	101.7	82.6	86.9	96.1	98.5	102.8
ALL4	39.0	38.0	35.2	157.1	19.6	61.9	58.9	66.1	71.6	72.8	76.0	70.3	70.2	71.8
ALL_AML	28.7	28.7	27.3	100.6	14.7	39.9	32.6	41.0	42.7	42.7	44.1	43.5	42.5	43.3
Adenoma	28.3	27.7	25.7	100.8	12.9	28.6	25.6	29.4	29.9	37.5	38.0	32.1	29.5	30.3
BASEHOCK	130.1	147.8	221.5	369.1	353.2	644.7	616.0	749.7	787.4	397.0	462.4	714.5	739.5	771.9
COIL20	68.3	64.3	69.7	63.2	37.7	106.0	91.6	111.6	114.2	125.4	129.6	102.6	104.3	111.8
Colon	18.1	18.5	15.6	36.3	12.2	23.0	22.8	23.1	23.5	26.7	26.8	26.3	24.2	24.8
DLBCL	28.3	28.4	27.1	98.5	15.8	40.1	36.0	42.3	44.3	46.1	48.9	44.2	43.8	44.9
Gastric	55.8	54.2	52.1	269.3	20.4	73.0	63.2	80.6	87.3	93.2	96.8	85.5	84.9	87.4
Gastric1	62.9	65.7	72.7	312.0	32.4	156.7	130.9	169.7	185.6	160.3	174.8	180.5	180.9	183.7
Gastric2	60.8	60.6	66.1	301.2	27.2	128.3	104.5	140.5	151.0	137.5	148.3	148.0	152.1	150.3
ISOLET	59.9	61.3	55.2	50.8	41.3	88.3	82.8	91.2	94.0	98.9	101.6	87.2	84.9	90.4
Leukaemia	28.8	28.8	27.4	98.3	14.4	39.2	32.7	40.9	42.4	41.5	43.5	43.3	43.1	43.0
Lymphoma	21.6	21.8	19.4	59.9	12.6	24.6	24.3	25.2	25.6	30.7	31.0	28.3	26.1	26.7
Myeloma	44.2	44.3	40.9	173.2	33.2	111.7	100.2	122.8	135.0	102.1	111.5	129.6	132.8	134.4
PCMAC	88.2	122.5	145.8	215.9	195.5	373.1	344.5	434.7	475.3	264.7	291.2	400.0	423.2	441.0
Prostate	39.6	38.9	38.4	168.2	24.5	65.3	60.5	69.8	76.5	73.8	77.7	74.0	74.4	74.7
SRBCT	23.0	22.4	19.5	41.2	13.4	25.3	25.1	25.6	26.0	31.8	32.3	28.8	26.8	27.4
T1D	118.8	117.4	113.2	663.6	58.3	256.6	216.1	283.6	315.2	265.7	296.1	311.4	312.2	307.8
madelon	41.5	40.4	40.1	45.3	34.1	81.4	73.7	89.0	94.2	76.5	80.4	71.8	75.8	86.9

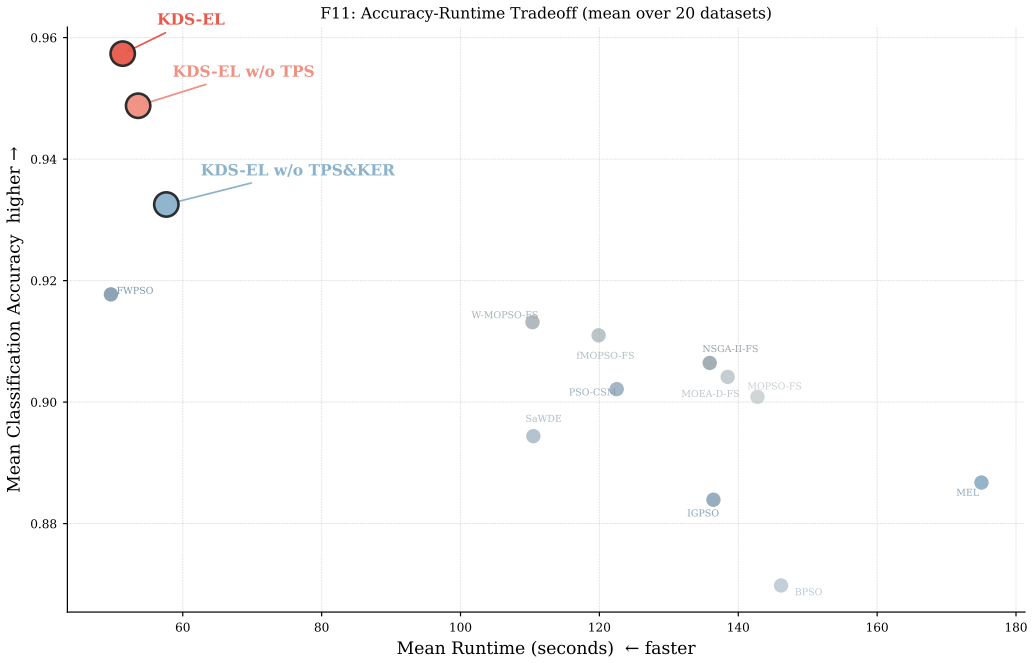


Figure 2: Classification accuracy versus runtime scatter plot across all 14 compared algorithms.

Section S6: Full Multi-Dataset Parameter Sensitivity Evaluation Grids (Table S6)

Table 11 reports the full multi-dataset one-factor-at-a-time (OFAT) parameter sensitivity evaluation results across the four representative benchmark datasets (*Adenoma*, *Colon*, *ALL4*, and *BASEHOCK*) for each of the five investigated hyper-parameters (WeightDecay, PhaseSplit, flip_frac, MI_init_ratio, and elite_k).

Table 11: Full parameter sensitivity results (mean accuracy, feature count, and runtime over the four datasets) for each of the five parameters. Bold rows indicate the default configuration.

Value	Avg Acc	Avg Feat	Avg Time (s)
WeightDecay			
0.90	0.932	158	49.2
0.95	0.939	127	50.8
0.99 (default)	**0.944**	**107**	**51.3**
0.995	0.943	112	52.1
1.00	0.941	151	53.7
PhaseSplit			
0.3	0.940	189	52.8
0.5	0.942	146	52.0
0.6	0.943	124	51.5
0.7 (default)	**0.944**	**107**	**51.3**
0.8	0.943	98	50.9
0.9	0.937	94	50.1
flip_frac			
0.005	0.939	96	54.8
0.01 (default)	**0.944**	**107**	**51.3**
0.02	0.942	118	50.6
0.05	0.937	156	49.9
0.10	0.931	203	48.7
MI_init_ratio			
0.0	0.932	184	62.3
0.25	0.940	137	55.8
0.5 (default)	**0.944**	**107**	**51.3**
0.75	0.942	101	49.6
1.0	0.936	96	47.8
elite_k			
1	0.938	124	48.9
2	0.942	115	50.2
3 (default)	**0.944**	**107**	**51.3**
5	0.941	118	52.7
10	0.935	141	55.9

Section S7: Full Convergence Trajectories and Boxplot Distributions

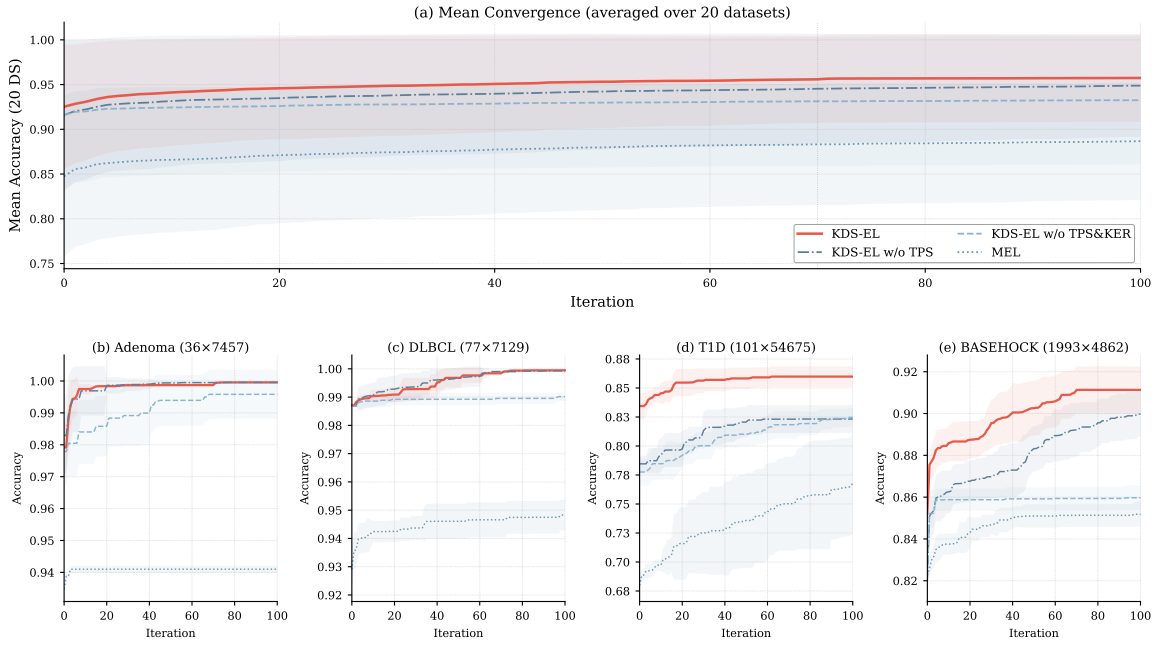


Figure 3: Evolutionary convergence trajectories across benchmark datasets comparing KDS-EL against baseline and competitive evolutionary algorithms.

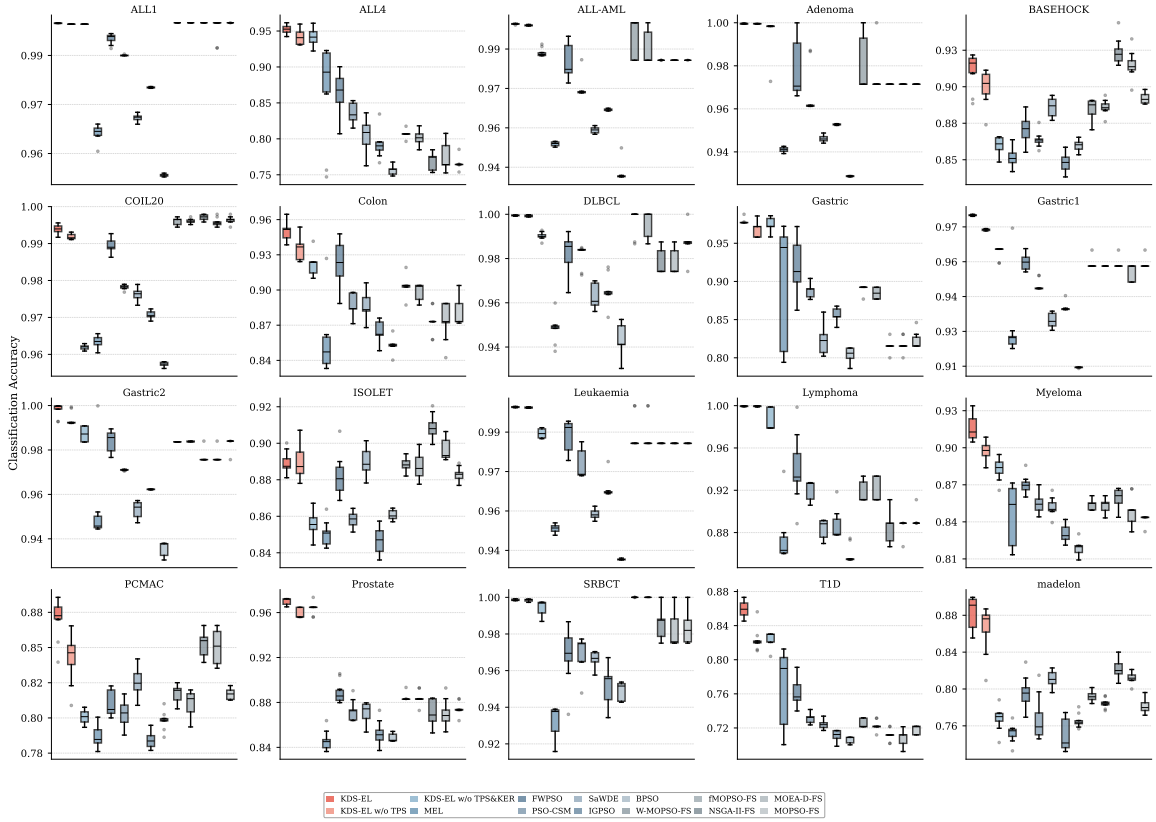


Figure 4: Boxplot distributions of classification accuracy across 10 independent runs for all compared algorithms across the benchmark suite.

Section S8: Extended Statistical Significance Testing Matrices

Table 12 and Table 13 summarize the pairwise non-parametric Wilcoxon signed-rank test outcomes and the Holm–Bonferroni step-down hypothesis rejection results.

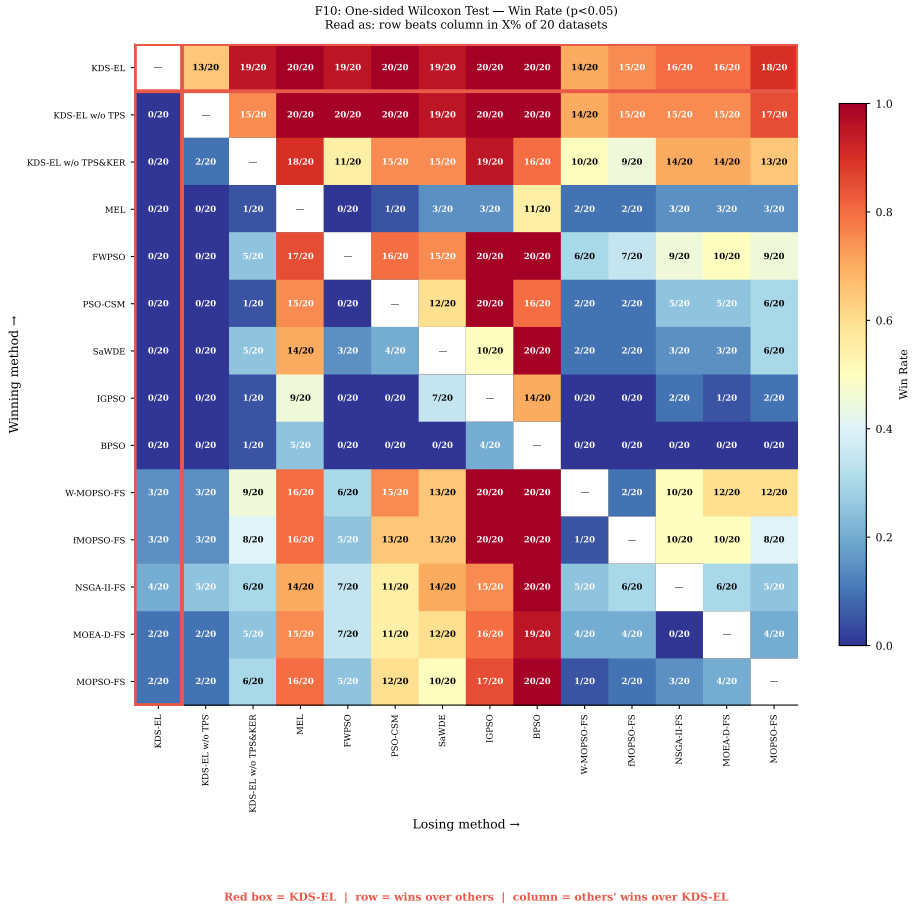


Figure 5: Pairwise Wilcoxon signed-rank test p-value heatmap across all evaluated algorithms.

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Table 12: One-sided Wilcoxon signed-rank test results ($p < 0.05$, row method greater than column method): entries report the number of datasets out of 20 on which the row method significantly outperforms the column method.

Method	KDS-EL	KDS-EL w/o TPS	KDS-EL w/o TPS&KER	MEL	FWPSO	PSO-CSM	SaWDE	IGPSO	BPSO	W-MOPSO-FS	fMOPSO-FS	NSGA-II-FS	MOEA-D-FS	MOPSO-FS
KDS-EL	—	13/20	19/20	20/20	19/20	20/20	19/20	20/20	20/20	14/20	15/20	16/20	16/20	18/20
KDS-EL w/o TPS	**0/20**	—	15/20	20/20	20/20	20/20	19/20	20/20	20/20	14/20	15/20	15/20	15/20	17/20
KDS-EL w/o TPS&KER	**0/20**	0/20	—	18/20	11/20	15/20	15/20	19/20	16/20	10/20	9/20	14/20	14/20	13/20
MEL	**0/20**	0/20	0/20	—	0/20	1/20	3/20	3/20	11/20	2/20	2/20	3/20	3/20	3/20
FWPSO	**0/20**	0/20	0/20	0/20	—	16/20	15/20	20/20	20/20	6/20	7/20	9/20	10/20	9/20
PSO-CSM	**0/20**	0/20	0/20	0/20	0/20	—	12/20	20/20	16/20	2/20	2/20	5/20	5/20	6/20
SaWDE	**0/20**	0/20	0/20	0/20	0/20	0/20	—	10/20	20/20	2/20	2/20	3/20	3/20	6/20
IGPSO	**0/20**	0/20	0/20	0/20	0/20	0/20	0/20	—	14/20	0/20	0/20	2/20	1/20	2/20
BPSO	**0/20**	0/20	0/20	0/20	0/20	0/20	0/20	0/20	—	0/20	0/20	0/20	0/20	0/20
W-MOPSO-FS	**0/20**	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	—	2/20	10/20	12/20	12/20
fMOPSO-FS	**0/20**	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	—	10/20	10/20	8/20
NSGA-II-FS	**0/20**	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	—	6/20	5/20
MOEA-D-FS	**0/20**	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	—	4/20
MOPSO-FS	**0/20**	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	—

Table 13: Holm–Bonferroni corrected pairwise comparison between KDS-EL and each of the 13 comparison methods over the 20 datasets. ΔAcc is the mean accuracy advantage of KDS-EL; Combined p is the unadjusted Fisher combined one-sided Wilcoxon p-value and Holm p is its Holm–Bonferroni adjustment; H_0 of no superiority is rejected at the Holm threshold in every case.

Method	ΔAcc	Combined p	Holm p	Reject H_0
KDS-EL w/o TPS	+0.0086	$< 10^{-15}$	$< 10^{-15}$	✓
KDS-EL w/o TPS&KER	+0.0248	$< 10^{-15}$	$< 10^{-15}$	✓
MEL	+0.0706	$< 10^{-15}$	$< 10^{-15}$	✓
FWPSO	+0.0396	$< 10^{-15}$	$< 10^{-15}$	✓
PSO-CSM	+0.0552	$< 10^{-15}$	$< 10^{-15}$	✓
SaWDE	+0.0630	$< 10^{-15}$	$< 10^{-15}$	✓
IGPSO	+0.0734	$< 10^{-15}$	$< 10^{-15}$	✓
BPSO	+0.0875	$< 10^{-15}$	$< 10^{-15}$	✓
W-MOPSO-FS	+0.0442	$< 10^{-15}$	$< 10^{-15}$	✓
fMOPSO-FS	+0.0464	$< 10^{-15}$	$< 10^{-15}$	✓
NSGA-II-FS	+0.0509	$< 10^{-15}$	$< 10^{-15}$	✓
MOEA-D-FS	+0.0532	$< 10^{-15}$	$< 10^{-15}$	✓
MOPSO-FS	+0.0565	$< 10^{-15}$	$< 10^{-15}$	✓

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